NUCLEAR FUEL CYCLE

High-Assay Low-Enriched Uranium (HALEU)

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The current fleet of nuclear reactors runs primarily on uranium fuel enriched up to 5% uranium-235 (U-235). High-assay low-enriched uranium (HALEU) is defined as uranium enriched to greater than 5% and less than 20% of the U-235 isotope. Applications for HALEU are today limited to research reactors and medical isotope production. However, HALEU will be needed for many advanced power reactor fuels, and more than half of the small modular reactor (SMR) designs in development.

HALEU is not yet widely available commercially. At present only Russia and China have the infrastructure to produce HALEU at scale, though the USA enacted a ban on Russian uranium imports in 2024 and is investing heavily in building domestic HALEU supply capacity. In the USA, Centrus Energy had produced over 920 kg of HALEU from a demonstration cascade at Piketon, Ohio, between October 2023 and mid-2025.

HALEU can be produced with existing centrifuge technology but requires a specific nuclear fuel cycle infrastructure and the development of new or modified regulations and licensing regimes. Moreover, new or modified transport containers will be required for the movement of the large quantities of HALEU required for the deployment of SMRs and advanced reactors.

Establishing the supply chain to produce and deliver HALEU to customers will require significant capital investment. Governments are playing a key role: in January 2024 the UK announced it had allocated £300 million to support HALEU production; and in January 2026 the US Department of Energy committed $2.7 billion over ten years to expand domestic uranium enrichment capacity.

Privacy - Terms

What is HALEU?

HALEU is defined as uranium enriched to greater than 5% and less than 20% of the U-235 isotope.

Uranium found in nature consists largely of two isotopes, U-235 and U-238. The production of energy in nuclear reactors is from the 'fission' or splitting of U-235 atoms, a process which releases energy in the form of heat. U-235 is the main fissile isotope of uranium.

Natural uranium contains 0.7% of the U-235 isotope. The remaining 99.3% is mostly the U-238 isotope which does not contribute directly to the fission process. Most reactors in operation today are light water reactors (of two types – PWR and BWR) and require uranium to be enriched from 0.7% to 3-5% U-235 in their fuel. HALEU is needed for many advanced power reactor fuels, and about two-thirds of the SMR designs in development.

Table 1: LEU, LEU+ and HALEU

	Low-enriched uranium			
		LEU+	HALEU	
Enrichment assay	<5%	5%-10%	5%-19.75%	19.75%
Chemical form	Oxide	Oxide	Oxide, salt, metal	Metal
End use	Generation III	Generation III	Generation IV (inc. SMRs)	Research reactors

Demand for HALEU

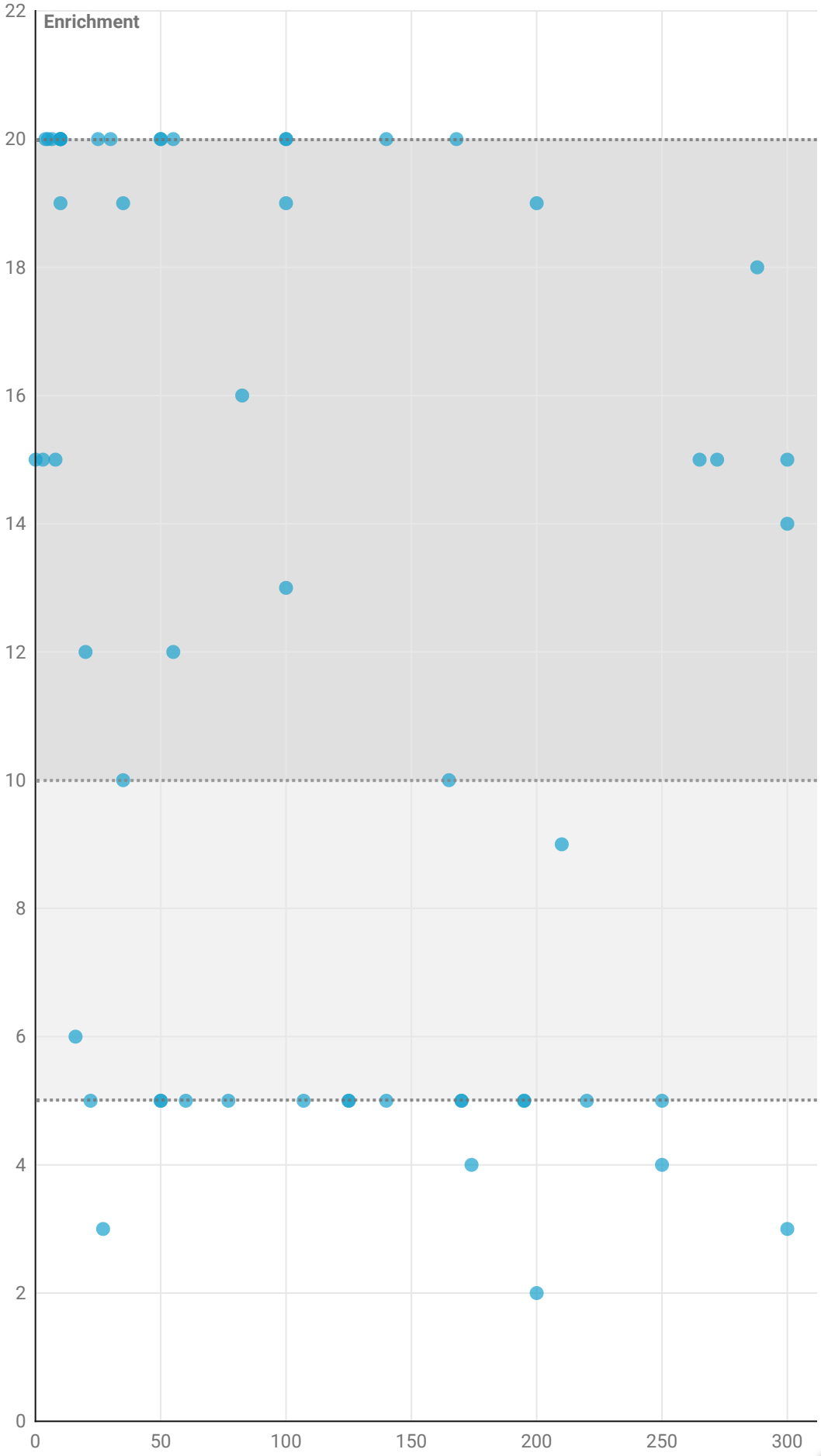
In the early stages of development of the nuclear industry, the process used to enrich uranium (gaseous diffusion) was energy-intensive and costly. Largely for economic reasons, therefore, the commercial nuclear energy industry has traditionally operated reactors with fuels enriched up to 5% U-235.

The technology transition to centrifugation-based enrichment significantly decreased production costs and enhanced the economic attractiveness of more highly enriched uranium. More highly enriched uranium requires more energy to produce, but allows for longer operating cycles, smaller reactors, and reduced production of radioactive waste.

Demand for HALEU is expected to increase going forward for three reasons:

- Use of 5-10% HALEU in existing conventional light water reactors.
- Use of 10-20% HALEU in advanced reactors and SMRs
- Ongoing transition, which started in 1990s, of research reactors from high-enriched uranium (HEU) to HALEU.

Figure 1: SMR designs, fuel enrichment vs. gross power output



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Box 1: Cost-benefit consideration of using HALEU

Potential benefits

- increased fuel burnup, increased capacity factor
- smaller reactor cores and reactors
- longer core lives / refueling cycles
- reduced waste volumes

Potential costs

- higher specific fuel costs (more energy required for enrichment)
- higher fuel fabrication costs to mitigate increased internal pressures, cladding corrosion etc.
- potential for accelerated corrosion and embrittlement of pressure vessels – potentially life-limiting reactor component
- potentially more onerous regulatory requirements and transport standards

The balance of these considerations will change based on the cost of uranium and fuel cycle services, and in respect of technological and regulatory considerations.

Producing HALEU

Enrichment

HALEU can be produced using existing centrifuge technology. Feed material for a HALEU cascade is typically uranium enriched to 4.95% U-235[1]. The feed material can be produced onsite by an adjacent cascade or be purchased and transported to site. About 85% of the SWU needed to produce HALEU is already contained in the feed material.

At present only Russia and China have the infrastructure to produce HALEU at scale. Commercial supply of HALEU is only available from Russian company Tenex. One company in the United States, Centrus Energy, began operating a pilot HALEU cascade in October 2023[2]. The USA enacted a ban on imports of Russian uranium products in May 2024, and Tenex is no longer able to supply the US market. In the USA, Centrus Energy began operating a demonstration HALEU cascade at Piketon, Ohio in October 2023 and by mid-2025 had produced and delivered over 920 kg of HALEU to the US Department of Energy. Several companies are developing commercial HALEU enrichment capacity in the West: Urenco received NRC authorization to enrich up to 10% U-235 in the USA in 2025, and is building a HALEU facility at Capenhurst in the UK with government support, targeting production by 2031. Orano is developing a new centrifuge enrichment facility (Project IKE) at Oak Ridge, Tennessee, with DOE support.

Downblending

HALEU can also be made from downblending HEU. The US Department of Energy is downblending surplus HEU at the Savannah River Site in South Carolina, where about 2.2 tonnes of HEU is expected to yield approximately 3.1 tonnes of HALEU over two-to-four years. This would provide an initial source of HALEU for advanced reactor developers until commercial enrichment capacity is established.

Security implications

Nuclear security is the responsibility of each state, but the IAEA provides recommendations and guidance. The IAEA defines three categories of nuclear material which provide the basis for states to specify the physical protection measures necessary at facilities storing or handling the material, and during its transport[3]. The principal factor that determines the necessary physical security protection measures for nuclear material is the potential for the material to be used in an explosive device.

Uranium used in today's nuclear reactors is a Category III material irrespective of amount. HALEU in the quantities required by commercial nuclear power plants will constitute a Category II material. This classification will have physical security implications

across the fuel cycle, for example at enrichment plants and at fuel fabrication facilities, as well as during transport.

Table 2: Categorization of nuclear materials (source: IAEA)

Material	Form	Category I	Category II	Category III[c]
1. Plutonium[a], unirradiated[b]		2kg or more	Less than 2kg but more than 500g	500g or less but more than 15g
2. Uranium-235, unirradiated[b]	U enriched to 20% U-235 or more	5kg or more	Less than 5kg but more than 1kg	1kg or less but more than 15g
	U enriched to 10% U-235 but less than 20%	-	10kg or more	Less than 10kg but more than 1kg
	U enriched above natural U-235 but less than 10%	-	-	10kg or more
3. Uranium-233, unirradiated[b]	2kg or more	2kg or more	Less than 2kg but more than 500g	500g or less but more than 15g
4. Irradiated fuel			Depleted or natural uranium, thorium or low enriched fuel (less than 10% fissile content).[d, e]	

a. All plutonium except that with isotopic concentration exceeding 80% in plutonium-238
b. Material not irradiated in a reactor or material irradiated in a reactor with a radiation level equal to or less than 1 Gy/h (100 rad/h) at 1m unshielded.
c. Quantities not falling in Category III and natural uranium, depleted uranium and thorium should be protected at least in accordance with prudent management practice. Although this level of protection is recommended, it would be open to States, upon evaluation of the specific circumstances, to assign a different category of physical protection.
d. Other fuel which by virtue of its original fissile material content is classified as Category I or II before irradiation may be reduced one category level while the radiation level from the fuel exceeds 1 Gy/h (100 rad/h) at one metre unshielded.

Facilities

In the United States the NRC defines facilities storing or processing category III quantities of nuclear material as being of 'low strategic importance' whilst facilities processing category II quantities of nuclear material are deemed to be of 'moderate strategic importance.' Security measures are site-specific, but the NRC has said that supplemental security measures for category II facilities could include: greater security or control over material in use and storage and vital equipment, requirements for access controls (e.g. background checks); controlled access area (CAA) portals and vehicle access, escort requirements, random entry searches and exit searches, alarm stations, security patrols, communication and coordination with law enforcement, and implementation of a security equipment maintenance program[4].

It is expected that the additional capital investments needed to build, license and operate such facilities will be significant.

Transport

Security implications of the widespread use of HALEU extend to transportation, as outlined in the Convention on the Physical Protection of Nuclear Material.

For example, category II material is to be protected with armed guards and escorts in certain circumstances, such as during storage incidental to nuclear transport (see below). Whilst category II material is transported internationally without issue, it is expected that additional security measures may be needed for transporting the sort of quantities required for a fleet of advanced reactors.

Safeguards implications

The aim of traditional IAEA safeguards is to detect the diversion of significant quantities of nuclear material from peaceful use in a timely manner. The IAEA has experience in safeguarding HALEU material in research reactors, but not as a commercial fuel.

Whilst IAEA security guidance (see above) distinguishes between LEU of different enrichment levels, its safeguards system does not. The IAEA uses two categories for its safeguards system for

enriched uranium: LEU and HEU. The IAEA considers LEU, uranium of less than 20% U-235, to be an 'indirect material' insofar as it cannot be used to directly manufacture nuclear weapons. HEU, uranium of 20% or more U-235, is considered a direct use material. The categories are used to determine the frequency of inspection and safeguards activities needed.

Facilities are inspected based on defined significant quantities (SQ) and material conversion times of nuclear materials. For LEU, the SQ is estimated to be 75kg of U-235 with a material conversion time of 3-12 months. For HEU, the SQ is estimated to be 25kg of U-235 with a material conversion time of 1-3 months[5].

SQ = the approximate amount of nuclear material for which the possibility of manufacturing a nuclear explosive device cannot be excluded.

Material conversion time = the time required to convert different forms of nuclear material into the metallic components of a nuclear explosive device.

The higher enrichment level of HALEU means less work is required to turn it into weapons-grade HEU[6]. In practice this means:

- The time required to produce weapons-grade HEU is shorter when using HALEU as feedstock
- A significant quantity (SQ) can be obtained from a relatively small amount of HALEU versus less highly enriched uranium.
- Facilities to produce weapons-grade HEU from HALEU feedstock could be relatively small and so difficult to detect.

It is expected therefore that the widespread deployment of advanced reactors using HALEU will affect the implementation of safeguards by the IAEA. The IAEA has significant scope to adapt its approach for HALEU facilities within its existing safeguards framework, as set out in INFCIRC/1537[7]. For example, the maximum routine inspection effort is proportional to a facility's inventory or annual throughput and the enrichment of the uranium, for facilities that contain uranium above 5% enrichment.

Safety implications

Criticality is a key consideration in the licensing of facilities or transportation packages handling fissile material. Required safety margins are determined using computer analyses benchmarked

against critical experiments applicable to the facility or package being licensed[8].

HALEU facilities or transportation packages will require wider criticality safety margins than those involved in the fuel cycle of uranium enriched to <5%, likely increasing costs. In the short term, an additional challenge is the lack of criticality benchmark data available for use[9].

Transport

Material used in commercial nuclear reactors is transported several times during the fuel cycle. Transport is regulated to protect people and the environment from the effects of radiation both routinely and when transport accidents occur. The fundamental principle is that the protection comes from the design of the package, regardless of how the material is transported.

More specifically, protection is achieved by:

- Containment of radioactive contents.
- Control of external radiation levels.
- Prevention of criticality.
- Prevention of damage caused by heat

The approved payloads of casks decrease with increased enrichment, presenting an economic challenge for the commercial transport of HALEU. For example, in the United States type 8A casks, which were designed to transport UF6 with an enrichment of up to 12.5%, are limited to 255 lbs. Type 5A and 5B casks can transport UF6 enriched up to 100%, but are limited to a payload of 54.9 lbs. At present there are no approved casks that would permit the economical transport of HALEU in the quantities required to sustain a commercial fuel cycle. In 2025, the US Department of Energy awarded $11 million to five companies to develop and license new HALEU transport packages.

'Chicken and egg' problem

A HALEU fuel cycle will incorporate new enrichment facilities, transportation solutions, and conversion and deconversion facilities.

The main challenge in establishing the necessary infrastructure is the lack of assured long-term commercial demand for HALEU. Without a clear demand signal private fuel cycle companies cannot commit the required capital. This 'chicken and egg' problem has stalled the build out of HALEU infrastructure and threatens to delay the deployment of advanced reactors and SMRs. Recent government investment in the USA and UK has begun to address this.

Notes & references

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